EDS 212: Day 2, Lecture 1

The exponential function, logarithms, and limits

August 4th, 2025

Overview of Day 2 topics

- The exponential function
- Logarithms
- Limits
- Derivatives

The exponential function

The number e

The **number e is a constant** in mathematics, it is approximately $e \approx 2.72$, but really its decimals continue infinitely without any pattern:

$$e = 2.718281828459045235 36...$$

It's exact value can be described as

$$e = 1 + 1 + \frac{1}{1 * 2} + \frac{1}{1 * 2 * 3} + \frac{1}{1 * 2 * 3 * 4} + \frac{1}{1 * 2 * 3 * 4 * 5} + \ldots$$

Like any other number, e follows normal power properties:

$$\begin{aligned} e^x \cdot e^y &= \qquad ? \\ \frac{1}{e^x} &= \qquad ? \\ \frac{e^x}{e^y} &= \qquad ? \\ (e^x)^r &= \qquad ? \end{aligned}$$

✏️ Take a moment to write these with a single exponent.

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Like any other number, e follows normal power properties:

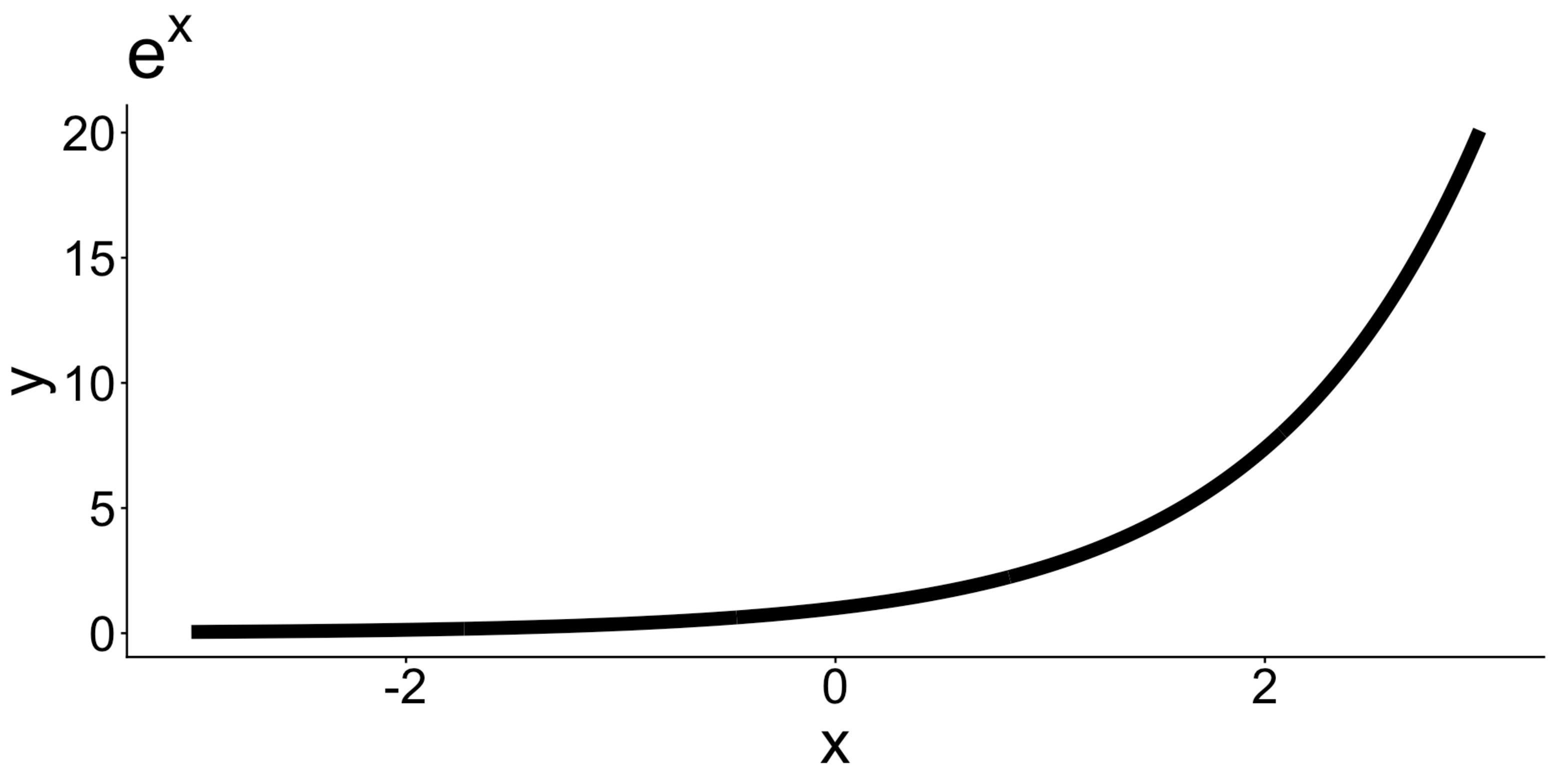
$$\begin{aligned} e^{x+y} &= e^x \cdot e^y \\ e^{-x} &= \frac{1}{e^x} \\ e^{x-y} &= \frac{e^x}{e^y} \\ e^{rx} &= (e^x)^r \end{aligned}$$

The exponential funciton

The **exponential function** e^x is the number e raised to the x power.

In function notation you may sometimes see it as

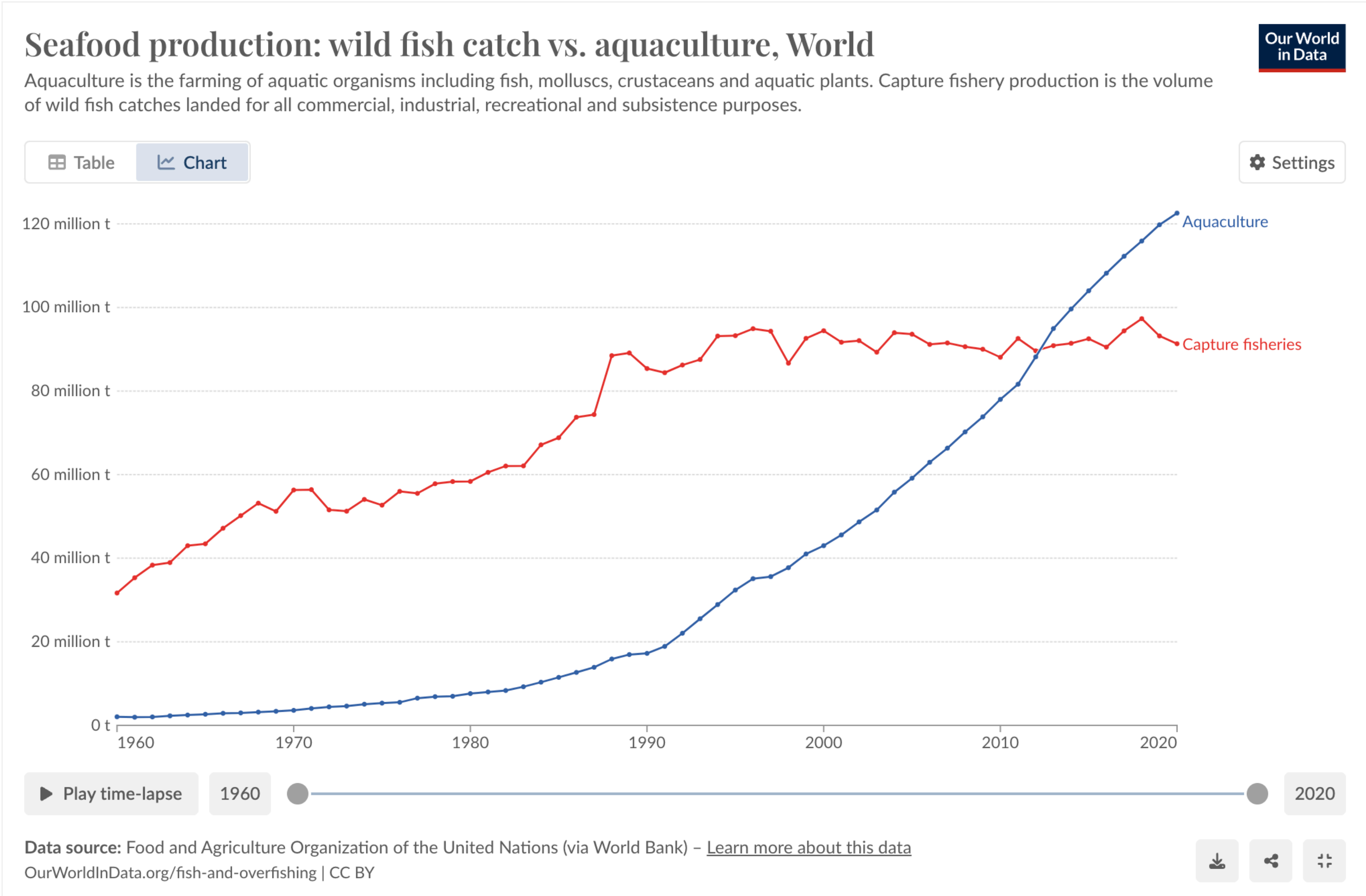
$$f(x) = e^x \quad \text{or} \quad \exp(x).$$



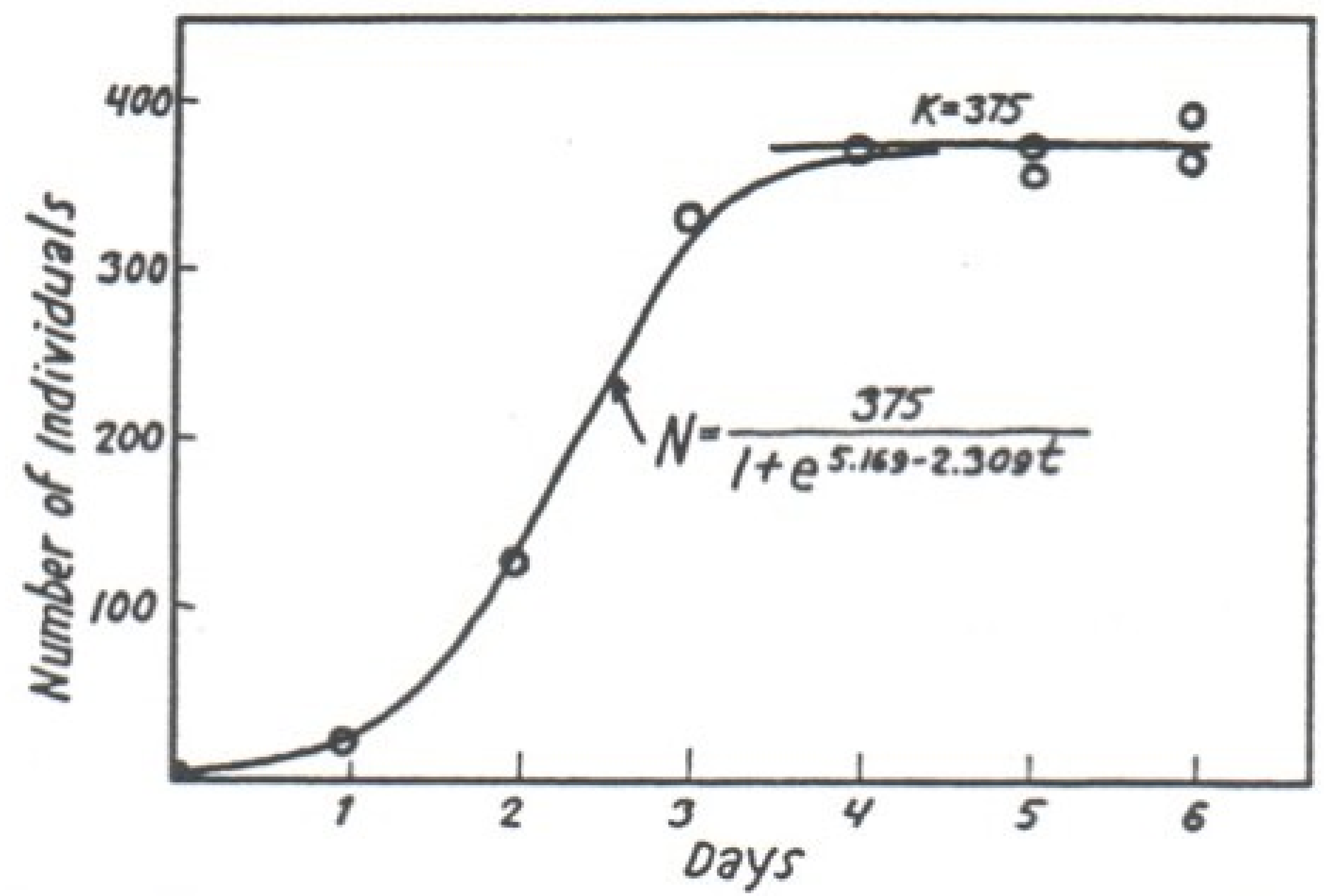
Why is the exponential function so common?

- **One reason:** Exponential trends show up a LOT in environmental science (the proportional change is the same over each time span)
- **Math reason:** Turns out it's a very useful value for calculus

A real-world example:



But populations can't grow exponentially forever



The growth of population of *Paramecium caudatum*

Gause, G. F. 1934. The Struggle for Existence. Baltimore: Williams and Wilkins.

Logistic growth

$$N_t = \frac{K}{1 + [\frac{K - N_0}{N_0}]e^{-rt}}$$

Where N_t is the population size at time t, K is the carrying capacity, N_0 is the initial population size, and r is a growth rate.

✏️ What should be the value of the formula when $t = 0$? Confirm your answer by substituting.

We should always *understand the components of formulas and equations* - both conceptually and mathematically.

Logistic growth

$$N_t = \frac{K}{1 + [\frac{K - N_0}{N_0}]e^{-rt}}$$

1. Why might we expect logistic growth for many populations?
2. What variables *besides* time would influence the actual population?

Logarithms

Logarithms

$\log_a(b)$ is the exponent you need to raise a to get a value of b

For example:

- $\log_2(8) = x$ asks "to what power do I have to raise 2, to get a value of 8?"
- $\log_{105}(1) = x$ asks "to what power do I have to raise 105, to get a value of 1?"

Natural logarithms

Natural logarithms

If $y > 0$, then $\ln(x)$ **is the exponent you need to raise e to in order to get y.**

This means:

$$\ln(e^x) = x$$
$$e^{\ln(y)} = y$$

In other words, $\ln(x)$ is the inverse of the exponential function e^x.

✏️ **Examples**

- $\ln(e^3)$
- $\ln(\frac{1}{e})$
- $\ln(1)$
- $\ln(0)$
- $\ln(-7)$

Natural logarithms

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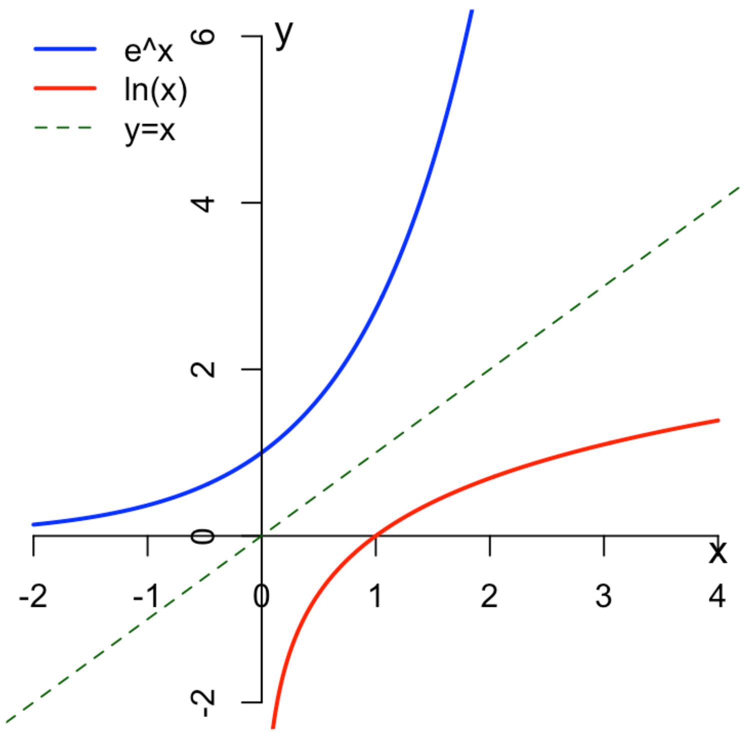
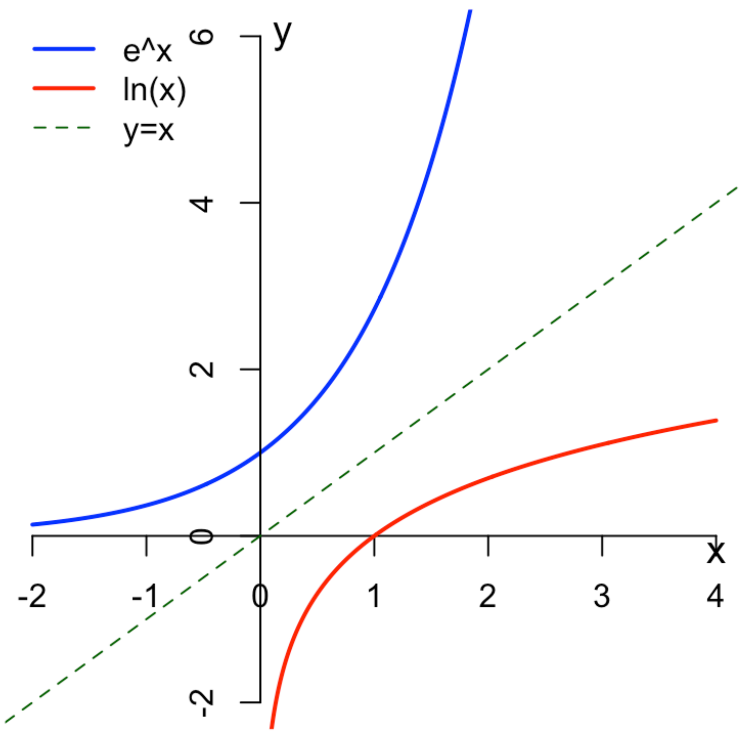
Examples

- $\ln(e^3) = 3$ because 3 is the power of e needed to get e^3.
- $\ln(\frac{1}{e}) = \ln(e^{-1}) = -1$
- $\ln(1) = 0$ because 0 is the exponent we need to raise e to to get 1
- $\ln(0)$ does not exist, because there's no number we can raise e to to get 0
- $\ln(-7)$ does not exist, because e to any number always be positive

Natural log properties



Natural log properties



Properties

$$\ln(e^x) = x$$

$$\ln(xy) = \ln x + \ln y$$

$$\ln(\frac{x}{y}) = \ln x - \ln y$$

$$\ln(x^y) = y \ln x$$

Using logarithms to solve equations

✏️ Solve for t in the equation $Pe^{rt} = A$.

Properties

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Using logarithms to solve equations

✏️ Solve for t in the equation $Pe^{rt} = A$.

$$\begin{aligned} Pe^{rt} &= A \\ e^{rt} &= \frac{A}{P} \\ rt &= \ln\left(\frac{A}{P}\right) \\ t &= \frac{\ln(A) - \ln(P)}{r} \end{aligned}$$

Limits

Limits

Definition

We say a number L is the limit of a function $f(x)$ as its variable x approaches a number c, if the function's output values $f(x)$ approach L when x get closer to c.

We write this symbolically as:

$$\lim_{x \to c} f(x) = L$$

We read this as: "The limit of $f(x)$ as x approaches c is L."

Limits example: numerical

What is the limit of $f(x) = 2x^2 - 4$ as x approaches 2?

We need to examine the values of $f(x)$ as x gets closer to 2 from both sides.

x approaches 2 from negative direction → ← x approaches 2 from positive direction

x	1.5	1.75	1.95	1.99	**2.0**	2.01	2.05	2.25	2.5
f(x)	0.5	2.12	3.60	3.92	**4.0**	4.08	4.41	6.13	8.5

f(x) approaches 4 from negative direction → ← f(x) approaches 4 from positive direction

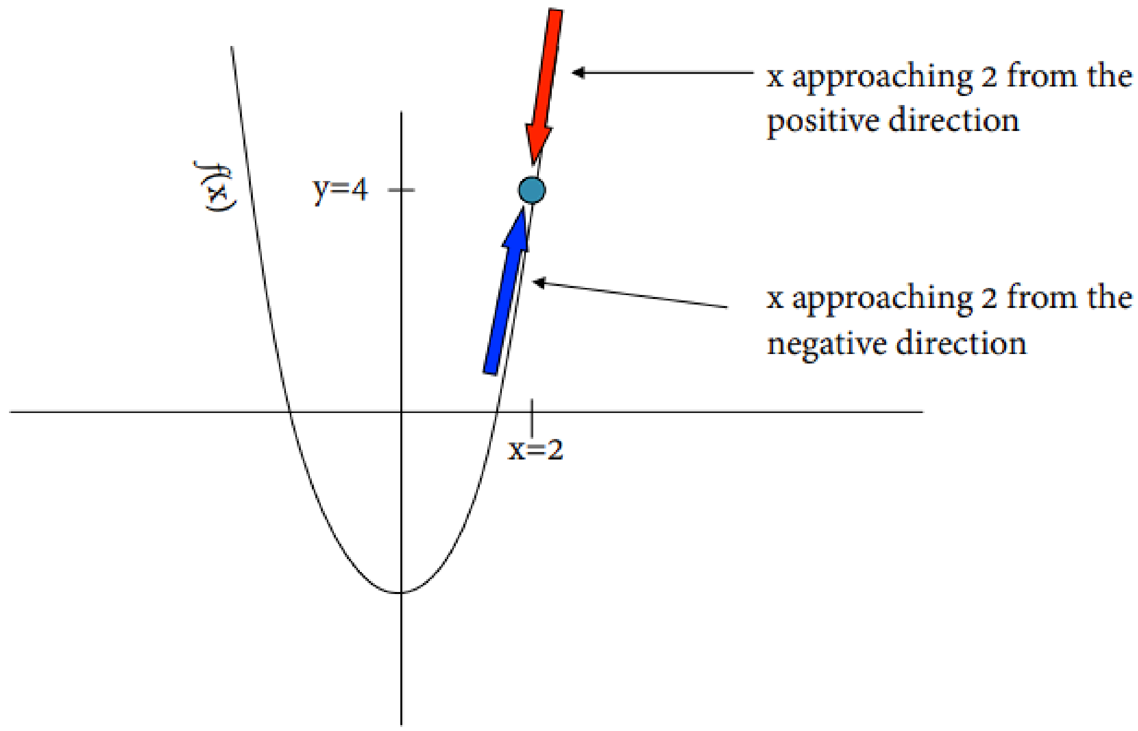
From both directions it looks like $f(x)$ converges to 4.

So, we say that "The limit of $2x^2 - 4$ as x approaches 2 is 4". We can write it like this:

$$\lim_{x \to 2}(2x^2 - 4) = 4$$

Limits example: graph

What is the limit of $f(x) = 2x^2 - 4$ as x approaches 2?



From both directions it looks like $f(x)$ converges to 4.

So, $\lim_{x\to 2}(2x^2 - 4) = 4$.

Do limits always exist?

The function $|x|$ is the **absolute value function**. It is such that

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}.$$

✏️ What happens to $f(x)$ at $x = 2$?

✏️ Try calculating:

$$\lim_{x \to 2} \frac{|x-2|}{x-2}.$$

Hint: Try to do numerical approximations or draw the graph of this function.

Do limits always exist?

✏️ What happens to $f(x)$ at $x = 2$?

This function is not defined at $x = 2$ since it would require us to divide by zero, which is undefined.

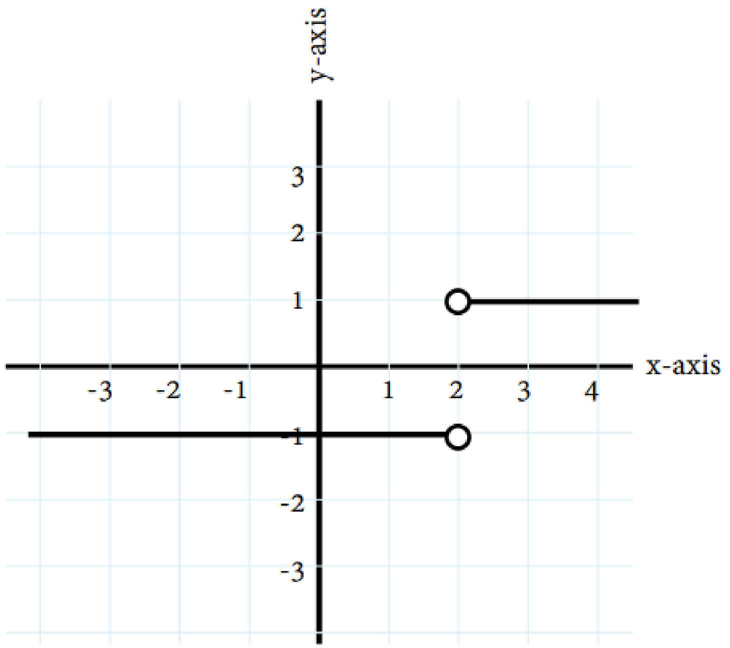
Try calculating $\lim_{x \to 2} \frac{|x-2|}{x-2}$.

Let's investigate:

x	1.9	1.99	1.999	2	2.001	2.01	2.1
f(x)	-1	-1	-1	?	1	1	1

Do limits always exist?

The function $\frac{|x-2|}{x-2}$ approaches different values from either side at $x = 2$.



Therefore... the limit does not exist!

Lunch break!

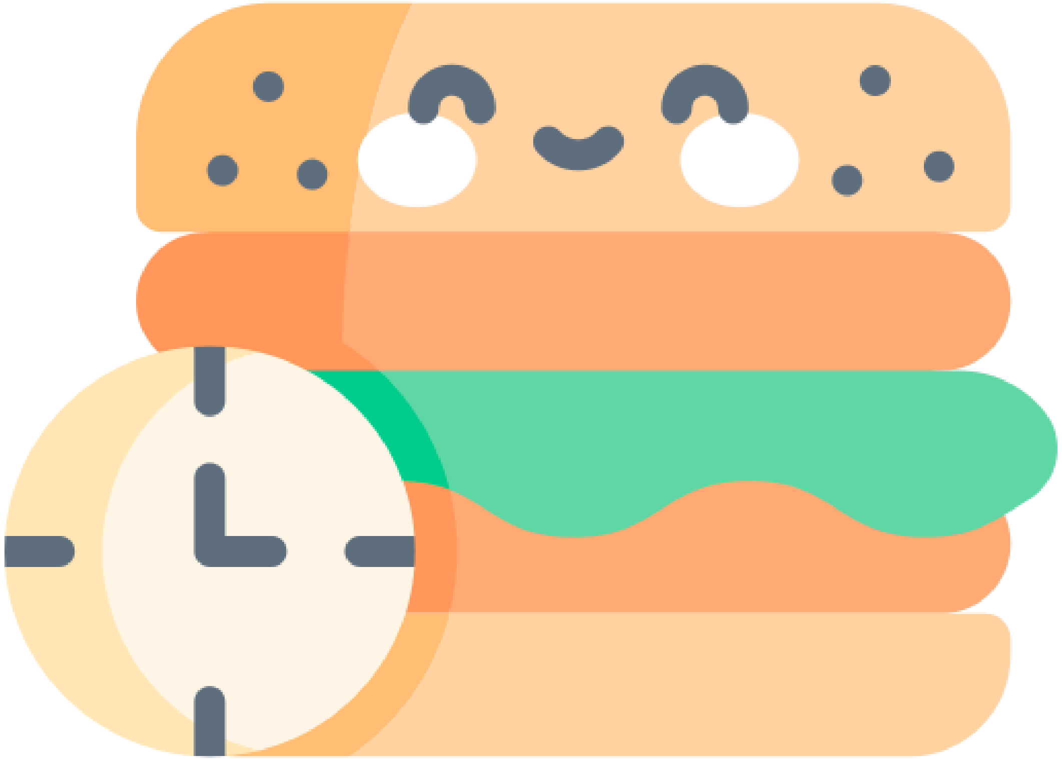


image: Flaticon.com